

Computer Science & Information Technology

Computer Networks

DPP: 1

Flow Control

- Q1** Consider two hosts are connected directly using point-to-point link. The link bandwidth is 10 Kbps, distance is 5 km, signal propagation speed is 10^5 meter per second and packet size is 200 bits than calculate transmission delay for a packet and one-way propagation delay (in millisecond) ?
(A) 20 and 50
(B) 20 and 500
(C) 200 and 50
(D) 200 and 500
- Q2** Consider two hosts S and D are connected directly using point-to-point link. The link bandwidth is 20 Kbps, signal propagation speed is 10^6 meter per second and distance is 40 km. Suppose host S wants to send a data of size 500 bits to host D in one packet. Calculate how much time (in millisecond) it takes to deliver the data completely from S to D ?
- Q3** Consider two hosts S and D are connected via one packet switch (takes negligible processing delay). Each link bandwidth is 1 Mbps, signal propagation speed is $4 * 10^6$ meter per second and length of each link is 20 km. Suppose host S wants to send a file of size 8000 bytes to host D, by dividing the file into 2000 bits packets. Calculate how much time (in millisecond) it takes to deliver the file completely from S to D ?
- Q4** Consider two hosts S and D are connected via two routers over a single path, routers takes negligible processing delay. Each link bandwidth is 100 Kbps, signal propagation speed is 10^6 meter per second and length of each link is 10 km. Suppose host S wants to send a file of size 5000 bytes to host D, by dividing the file into 1000 bits packets. Calculate how much time (in millisecond) it takes to deliver the file completely from S to D ?
- Q5** Consider two hosts are connected via point-to-point link using Stop-and-Wait protocol for flow control. The transmission rate of the link is 5 Kbps and the one-way propagation delay of the link is 50 milliseconds. The frame size is 500 bits. Calculate channel utilization in percentage (rounded off to nearest integer) ?
- Q6** Consider two hosts are connected via point-to-point link using Stop-and-Wait protocol for flow control. The transmission rate of the link is 2 Mbps and frame size is 4000 bytes. To achieve 50 percent link utilization, calculate one-way propagation delay in milliseconds ?
- Q7** Consider two hosts are connected via point-to-point link using Stop-and-Wait protocol for flow control. The transmission delay for a frame and one-way propagation delay of the link are 15 and 30 milliseconds respectively. If link bandwidth is 50 Kbps then calculate throughput of the link in Kbps ?
- Q8** Consider two hosts are connected via point-to-point link using Stop-and-Wait protocol for flow control. The link bandwidth is 4 Kbps, one-way propagation delay of the link are 800

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milliseconds. To achieve 50 percent link utilization calculate packet size in bytes ?

- Q9** Consider a packet error probability is 0.3 in Stop-and-Wait ARQ. What should be minimum number of transmission required to transmit 140 packets?
- Q10** Consider two hosts are connected via point-to-point link using Sliding Window protocol for flow control. The transmission delay for a frame and propagation delay of the link are 10 and 30 milliseconds. Calculate optimal window size ?
(A) 1 (B) 3
(C) 4 (D) 7
- Q11** Consider two hosts A and B are connected via point-to-point link using Go-Back-4-ARQ flow control strategy with transmit window size 4. Host A wants to send a file (divided into 10 packets) to host B, if every 6th packet that A transmits gets lost (but no ACKs from B ever get lost), then what is the number of packets that A will transmit for sending the file to B ?
- Q12** Consider two hosts connected directly through point-to-point link uses Go Back N ARQ for flow control. The transmission delay and propagation delay (in millisecond) are 2 and 250. What should be minimum number of bits required for sequence number field to achieve full utilization ?
(A) 7 (B) 8
(C) 9 (D) 10
- Q13** Consider a 2 Mbps communication link with one-way propagation delay of 120 milliseconds. Go back N ARQ protocol is used on this link to send data with a frame size of 2000 bits. Neglect the

transmission time of acknowledgment. The minimum number of bits required for the sequence number field to achieve 50% utilization is _____ .

- Q14** Consider two hosts connected directly through point-to-point link uses Selective Repeat ARQ for flow control. The transmission delay and propagation delay (in millisecond) are 4 and 250. What should be minimum number of bits required for sequence number field to achieve full utilization?
(A) 5 (B) 6
(C) 7 (D) 8
- Q15** Consider a 1 Mbps communication link with one-way propagation delay of 80 milliseconds. Selective repeat protocol is used on this link to send data with a frame size of 2000 bits. Neglect the transmission time of acknowledgment. The minimum number of bits required for the sequence number field to achieve 100% utilization is _____ .
- Q16** A bit-stuffing based framing protocol uses an 8-bit delimiter pattern of 01111110. If the input bit-string is 10111110111, then the output bit-string after stuffing is
(A) 10111110111 (B) 101111111
(C) 101111100111 (D) 1011111001110
- Q17** A bit-stuffing based framing protocol uses an 8-bit delimiter pattern of 01111110. If the output bit-string after stuffing is 10111110111, then the input bit-string is
(A) 10111110111 (B) 101111111
(C) 101111100111 (D) 1011111001110



Answer Key

Q1 (A)
Q2 65~65
Q3 76~76
Q4 450~450
Q5 50~50
Q6 8~8
Q7 10~10
Q8 800~800
Q9 200~200

Q10 (D)
Q11 17~17
Q12 (B)
Q13 7~7
Q14 (D)
Q15 8~8
Q16 (C)
Q17 (B)



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Hints & Solutions

Q1 Text Solution:

$$t_x = \frac{\text{Packet size}}{\text{Bandwidth}} = \frac{200 \text{ bits}}{10 \text{ Kbps}} = \frac{200 \text{ bits}}{10 \times 10^3 \text{ bits/sec.}}$$

$$= 20$$

$$t_p = \frac{\text{Distance}}{\text{Signal speed}} = \frac{5 \text{ KM}}{10^5 \text{ m/sec.}} = \frac{5 \times 10^3 \text{ sec.}}{10^5 \text{ m/sec.}}$$

$$= 50 \text{ ms}$$

Q2 Text Solution:

$$t_x = \frac{\text{Packet size}}{\text{Bandwidth}} = \frac{500 \text{ bits}}{20 \text{ Kbps}} = \frac{500 \text{ bits}}{20 \times 10^3 \text{ bits/sec.}}$$

$$= 25 \text{ ms}$$

$$t_p = \frac{\text{Distance}}{\text{Signal speed}} = \frac{40 \text{ KM}}{10^6 \text{ m/sec.}} = \frac{40 \times 10^3 \text{ sec.}}{10^6 \text{ m/sec.}}$$

$$= 40 \text{ ms}$$

$$\text{Total time} = (t_x + t_p) = (25 + 40) \text{ ms} = 65 \text{ ms}$$

Q3 Text Solution:

$$t_x = \frac{\text{Packet size}}{\text{Bandwidth}} = \frac{2000 \text{ bits}}{1 \text{ Mbps}} = 2 \text{ ms}$$

$$t_p = \frac{\text{Distance}}{\text{Signal speed}} = \frac{20 \text{ KM}}{4 \times 10^6 \text{ m/sec.}} = 5 \text{ ms}$$

$$\text{Number of packets}(N) = \frac{\text{File size}}{\text{Packet size}} = \frac{8000 \text{ bytes}}{2000 \text{ bits}} = 32$$

$$N = 32$$

$$\text{Total time} = (N \times t_x + t_p) + (t_x + t_p)$$

$$= (32 \times 2 + 5) + (2 + 5) \text{ ms}$$

$$= 76 \text{ ms}$$

Q4 Text Solution:

$$t_x = \frac{\text{Packet size}}{\text{Bandwidth}} = \frac{1000 \text{ bits}}{100 \text{ Kbps}} = 10 \text{ ms}$$

$$t_p = \frac{\text{Distance}}{\text{Signal speed}} = \frac{10 \text{ KM}}{10^6 \text{ m/sec.}} = 10 \text{ ms}$$

$$\text{Number of packets}(N) = \frac{\text{File size}}{\text{Packet size}} = \frac{5000 \text{ bytes}}{1000 \text{ bits}} = 40$$

$$\text{Total time} = (N \times t_x + t_p) + (t_x + t_p) + (t_x + t_p)$$

$$= (40 \times 10 + 10) + (10 + 10) + (10 + 10) \text{ ms}$$

$$= 450 \text{ ms}$$

Q5 Text Solution:

$$t_x = \frac{\text{Frame size}}{\text{Bandwidth}} = \frac{500 \text{ bits}}{5 \text{ Kbps}} = 100 \text{ ms}$$

$$t_p = 50 \text{ ms}$$

$$\text{Cycle time} = (t_x + 2t_p) = (100 + 2 \times 50) \text{ ms} = 200 \text{ ms}$$

$$\eta = \frac{t_x}{\text{cycle time}} \times 100\% = \frac{100 \text{ ms}}{200 \text{ ms}} \times 100\%$$

$$= 50\%$$

Q6 Text Solution:

$$t_x = \frac{\text{Frame size}}{\text{Bandwidth}} = \frac{4000 \text{ bytes}}{2 \text{ Mbps}} = \frac{32 \times 10^3 \text{ bits}}{2 \times 10^6 \text{ bits/sec.}}$$

$$= 16 \text{ ms}$$

Stop-and-wait protocol:-

$$\eta = \frac{t_x}{\text{cycle time}}$$

for 50% utilization

$$\text{cycle time} = 2t_x$$

$$(t_x + 2t_p) = 2t_x$$

$$t_p = \frac{t_x}{2} = \frac{16 \text{ ms}}{2} = 8 \text{ ms}$$

Q7 Text Solution:

$$t_x = 15 \text{ ms}, t_p = 30 \text{ ms}$$

$$\text{Bandwidth} = 50 \text{ Kbps}$$

$$\text{Cycle time} = (t_x + 2t_p) = (15 + 2 \times 30) \text{ ms} = 75 \text{ ms}$$

$$\eta = \frac{t_x}{\text{cycle time}} = \frac{15 \text{ ms}}{75 \text{ ms}} = \frac{1}{5}$$

$$\text{Throughput} = \eta \times \text{Bandwidth}$$

$$= \frac{1}{5} \times 50 \text{ Kbps} = 10 \text{ Kbps}$$

Q8 Text Solution:

For 50% utilization in stop-and-wait protocol

$$\text{cycle time} = 2t_x$$

$$(t_x + 2t_p) = 2t_x$$

$$t_x = 2t_p$$

$$\text{Frame size} = (2t_p) \times \text{Bandwidth}$$

$$= (2 \times 800 \text{ ms}) \times 4 \text{ Kbps}$$

$$= 64 \times 100 \times 10^{-3} \text{ sec.} \times 10^3 \text{ bits/sec.}$$

$$= 6400 \text{ bits}$$

$$= 800 \text{ bytes}$$

Q9 Text Solution:

$$\text{Number of Packets}(n) = 140$$

$$\text{Packet error probability}(P) = 0.3$$

$$\text{Total transmission required}(N) = ?$$



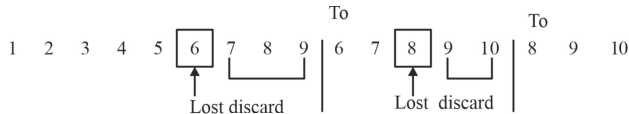
$$N = \frac{n}{1-P} = \frac{140}{(1-0.3)} = 200$$

Q10 Text Solution:

$$t_x = 10 \text{ ms}, t_p = 30 \text{ ms}$$

$$\text{cycle time} = (t_x + 2t_p) = (10 + 2 \times 30) \text{ ms} = 70 \text{ ms}$$

$$\text{Optimal window size} = \left\lceil \frac{\text{cycle time}}{t_x} \right\rceil = \left\lceil \frac{70 \text{ ms}}{10 \text{ ms}} \right\rceil = 7 \text{ ms}$$

Q11 Text Solution:**Q12 Text Solution:**

$$t_x = 2 \text{ ms}, t_p = 250 \text{ ms}$$

$$\text{cycle time} = (t_x + 2t_p)$$

$$= (2 + 2 \times 250) \text{ ms} = 502 \text{ ms}$$

$$\text{Optimal window size (N)} = \left\lceil \frac{\text{cycle time}}{t_x} \right\rceil$$

$$N = \left\lceil \frac{502 \text{ ms}}{2 \text{ ms}} \right\rceil = 251$$

$$\text{Minimum bit for sequence number} =$$

$$\lceil \log_2 (N + 1) \rceil \text{ bits}$$

$$= \lceil \log_2 (252) \rceil \text{ bits}$$

$$= 8 \text{ bits}$$

Q13 Text Solution:

$$t_x = \frac{\text{Frame size}}{\text{Bandwidth}} = \frac{2000 \text{ bits}}{2 \text{ M bps}} = 1 \text{ ms}$$

$$t_p = 120 \text{ ms}$$

$$\text{Cycle time} = (t_x + 2t_p) = (1 + 2 \times 120) \text{ ms} = 241 \text{ ms}$$

For 50% utilization

$$\text{Transmit window size (N)} =$$

$$\left\lceil \frac{\text{cycle time}}{2t_x} \right\rceil = \left\lceil \frac{241 \text{ ms}}{2 \times 1 \text{ ms}} \right\rceil = 121$$

$$\text{Minimum number of bit for sequence number} =$$

$$\lceil \log_2 (N + 1) \rceil \text{ bits}$$

$$= \lceil \log_2 (122) \rceil \text{ bits} = 7 \text{ bits}$$

Q14 Text Solution:

$$t_x = 4 \text{ ms}$$

$$t_p = 250 \text{ ms}$$

$$\text{Cycle time} = (t_x + 2t_p)$$

$$= (4 + 2 \times 250) \text{ ms} = 504 \text{ ms}$$

$$\text{Optimal window size (N)} =$$

$$\left\lceil \frac{\text{cycle time}}{t_x} \right\rceil = \left\lceil \frac{504 \text{ ms}}{4 \text{ ms}} \right\rceil = 126$$

$$N = 126$$

$$\text{Minimum bit for sequence number}$$

$$= \lceil \log_2 (2N) \rceil \text{ bits}$$

$$= \lceil \log_2 (252) \rceil \text{ bits}$$

$$8 \text{ bits}$$

Q15 Text Solution:

$$t_x = \frac{\text{Frame size}}{\text{Bandwidth}} = \frac{2000 \text{ bits}}{1 \text{ M bps}} = 2 \text{ ms}$$

$$t_p = 80 \text{ ms}$$

$$\text{cycle time} = (t_x + 2t_p) = (2 + 2 \times 80) \text{ ms} = 162 \text{ ms}$$

$$\text{Optimal window size (N)} =$$

$$\left\lceil \frac{\text{cycle time}}{t_x} \right\rceil = \left\lceil \frac{162 \text{ ms}}{2 \text{ ms}} \right\rceil = 81$$

$$\text{Minimum number of bits for sequence number}$$

$$= \lceil \log_2 (2N) \rceil \text{ bits}$$

$$= \lceil \log_2 (162) \rceil \text{ bits}$$

$$= 8 \text{ bits}$$

Q16 Text Solution:

$$\text{Frame delimiter} = 01111110$$

$$\text{Data} = 10111110111$$

$$\text{Transmitted data} = 101111100111$$

Q17 Text Solution:

$$\text{Frame Delimiter} = 01111110$$

$$\text{Transmitted Data} = 10111110111$$

$$\text{Data} = 101111111$$

